# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **How Quantities Combine (Understanding Data)** | Class 9 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 42 | Sum 210 divided by 5. |
| 2 | **(c)** 40 | Equal-sized groups: average of the averages. |
| 3 | **(b)** 87 | $29\%$ of $300 = \frac{29}{100} \times 300 = 87$. |
| 4 | **(a)** 95 | Total length is the sum of the components. |
| 5 | **(b)** 51 | $\frac{24 \times 46 + 15 \times 59}{39} = 51$. |
| 6 | **(c)** 30 mL | $\frac{12}{100} \times 250 = 30$. |
| 7 | **(c)** total of all values divided by the total number of values | Use totals, not the plain mean of the two averages. |
| 8 | **(b)** 72 | $\frac{79 \times 2 + 64 \times 1 + 69 \times 2}{5} = 72$. |
| 9 | **(d)** 33 | Sum 198 divided by 6. |
| 10 | **(c)** 43 | Equal-sized groups: average of the averages. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Sum $= 55 + 27 + 50 + 44 + 54 = 230$ (1); Mean $= \frac{230}{5} = 46$ (1). | 1+1 |
| 12 | Food $= \frac{25}{100} \times 200 = 50$; Remaining share $= 75\%$ (1); Everything else $= 200 - 50 = 150$ (1). | 1+1 |
| 13 | Pulp $= \frac{36}{100} \times 250 = 90$ mL (1); Remaining $= 250 - 90 = 160$ mL (1). | 1+1 |
| 14 | Sum $= 44 + 36 + 42 + 30 = 152$ (1); Mean $= \frac{152}{4} = 38$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Total of first group $= 10 \times 47 = 470$ (1); Total of second group $= 30 \times 31 = 930$ (1); Combined average $= \frac{1400}{40} = 35$ (1). | 1+1+1 |
| 16 | Sum of weekly averages $= 123$ (1); Average $= \frac{123}{3} = 41$ min per day (1); Every week has the same number of days, so each average has the same weight (1). | 1+1+1 |
| 17 | Weighted sum $= 83 \times 3 + 61 \times 3 = 432$ (1); Sum of weights $= 6$ (1); Weighted average $= 72$ (1). | 1+1+1 |
| | **OR:** Total of first group $= 24 \times 46 = 1104$ (1); Total of second group $= 15 \times 59 = 885$ (1); Combined average $= \frac{1989}{39} = 51$ (1). | 1+1+1 |
| 18 | Sum of weekly averages $= 225$ (1); Average $= \frac{225}{5} = 45$ min per day (1); Every week has the same number of days, so each average has the same weight (1). | 1+1+1 |
| | **OR:** Weighted sum $= 42 \times 2 + 58 \times 2 = 200$ (1); Sum of weights $= 4$ (1); Weighted average $= 50$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Sum of 12 observations $= 12 \times 50 = 600$ (1); New sum $= 600 + 63 = 663$ (2); New mean $= \frac{663}{13} = 51$ (2). | 1+2+2 |
| | **OR:** Pure part in the first $= 20 \times \frac{40}{100} = 8$ mL (1); Pure part in the second $= 16.8$ mL (1); Total $= 24.8$ mL in 80 mL (1); Concentration $= 31\%$ (2). | 1+1+1+2 |
| 20 | Sum of 10 observations $= 10 \times 35 = 350$ (1); New sum $= 350 + 57 = 407$ (2); New mean $= \frac{407}{11} = 37$ (2). | 1+2+2 |
| | **OR:** Pure part in the first $= 60 \times \frac{28}{100} = 16.8$ mL (1); Pure part in the second $= 4.8$ mL (1); Total $= 21.6$ mL in 80 mL (1); Concentration $= 27\%$ (2). | 1+1+1+2 |

**Mistakes to avoid:**

- Taking the combined average as $\frac{a_1 + a_2}{2}$ when the groups have different sizes.
- Adding the percentages of two solutions to find the mixture, or averaging the two concentrations without using the volumes.
- Dividing the sum of weighted scores by the number of items instead of by the sum of the weights.
- Forgetting to change the volume when water is added: the pure amount stays the same but the total increases.
- Using a percentage as if it were an amount (e.g. $20\%$ of 5 L is 1 L, not 20 L).
- Reading a 100% stacked bar as if the bar lengths showed the actual totals; the actual totals are not shown.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

## ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **How Quantities Combine (Understanding Data)** | Class 9 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 32 | $32\%$ of $100 = \frac{32}{100} \times 100 = 32$. |
| 2 | **(d)** 98 | Total length is the sum of the components. |
| 3 | **(a)** 42 | $\frac{24 \times 41 + 12 \times 44}{36} = 42$. |
| 4 | **(b)** 20 mL | $\frac{5}{100} \times 400 = 20$. |
| 5 | **(c)** the two groups have the same size | Equal weights. |
| 6 | **(a)** 67 | $\frac{85 \times 2 + 51 \times 3 + 79 \times 1}{6} = 67$. |
| 7 | **(c)** 24 | Sum 120 divided by 5. |
| 8 | **(c)** 40 | Equal-sized groups: average of the averages. |
| 9 | **(d)** 35 | $25\%$ of $140 = \frac{25}{100} \times 140 = 35$. |
| 10 | **(c)** 84 | Total length is the sum of the components. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Food $= \frac{30}{100} \times 140 = 42$; Remaining share $= 70\%$ (1); Everything else $= 140 - 42 = 98$ (1). | 1+1 |
| 12 | Pulp $= \frac{28}{100} \times 350 = 98$ mL (1); Remaining $= 350 - 98 = 252$ mL (1). | 1+1 |
| 13 | Sum $= 12 + 17 + 62 + 42 + 68 + 15 = 216$ (1); Mean $= \frac{216}{6} = 36$ (1). | 1+1 |
| 14 | Food $= \frac{15}{100} \times 160 = 24$; Remaining share $= 85\%$ (1); Everything else $= 160 - 24 = 136$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Total of first group $= 26 \times 39 = 1014$ (1); Total of second group $= 18 \times 61 = 1098$ (1); Combined average $= \frac{2112}{44} = 48$ (1). | 1+1+1 |
| 16 | Sum of weekly averages $= 208$ (1); Average $= \frac{208}{4} = 52$ min per day (1); Every week has the same number of days, so each average has the same weight (1). | 1+1+1 |
| 17 | Weighted sum $= 51 \times 2 + 95 \times 2 = 292$ (1); Sum of weights $= 4$ (1); Weighted average $= 73$ (1). | 1+1+1 |
| | **OR:** Total of first group $= 26 \times 61 = 1586$ (1); Total of second group $= 13 \times 58 = 754$ (1); Combined average $= \frac{2340}{39} = 60$ (1). | 1+1+1 |
| 18 | Sum of weekly averages $= 144$ (1); Average $= \frac{144}{3} = 48$ min per day (1); Every week has the same number of days, so each average has the same weight (1). | 1+1+1 |
| | **OR:** Weighted sum $= 95 \times 2 + 89 \times 2 = 368$ (1); Sum of weights $= 4$ (1); Weighted average $= 92$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Sum of 7 observations $= 7 \times 41 = 287$ (1); New sum $= 287 + 57 = 344$ (2); New mean $= \frac{344}{8} = 43$ (2). | 1+2+2 |
| | **OR:** Pure part in the first $= 40 \times \frac{18}{100} = 7.2$ mL (1); Pure part in the second $= 4.8$ mL (1); Total $= 12$ mL in 60 mL (1); Concentration $= 20\%$ (2). | 1+1+1+2 |
| 20 | Sum of 9 observations $= 9 \times 44 = 396$ (1); New sum $= 396 + 54 = 450$ (2); New mean $= \frac{450}{10} = 45$ (2). | 1+2+2 |
| | **OR:** Pure part in the first $= 40 \times \frac{27}{100} = 10.8$ mL (1); Pure part in the second $= 13.2$ mL (1); Total $= 24$ mL in 100 mL (1); Concentration $= 24\%$ (2). | 1+1+1+2 |

**Mistakes to avoid:**

- Taking the combined average as $\frac{a_1 + a_2}{2}$ when the groups have different sizes.

- Adding the percentages of two solutions to find the mixture, or averaging the two concentrations without using the volumes.
- Dividing the sum of weighted scores by the number of items instead of by the sum of the weights.
- Forgetting to change the volume when water is added: the pure amount stays the same but the total increases.
- Using a percentage as if it were an amount (e.g. 20% of 5 L is 1 L, not 20 L).
- Reading a 100% stacked bar as if the bar lengths showed the actual totals; the actual totals are not shown.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **How Quantities Combine (Understanding Data)** | Class 9 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** 48 | $\frac{24\times42+18\times56}{42}=48.$ |
| 2 | **(b)** 90 mL | $\frac{15}{100}\times600=90.$ |
| 3 | **(d)** the total of all the components | Whole bar = total. |
| 4 | **(d)** 79 | $\frac{72\times3+76\times2+88\times3}{8}=79.$ |
| 5 | **(c)** 37 | Sum 148 divided by 4. |
| 6 | **(a)** 30 | Equal-sized groups: average of the averages. |
| 7 | **(c)** 75 | $25\%$ of $300=\frac{25}{100}\times300=75.$ |
| 8 | **(a)** 80 | Total length is the sum of the components. |
| 9 | **(b)** 55 | $\frac{28\times59+14\times47}{42}=55.$ |
| 10 | **(a)** 280 g | $\frac{35}{100}\times800=280.$ |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Pulp $=\frac{19}{100}\times300=57$ mL (1); Remaining $=300-57=243$ mL (1). | 1+1 |
| 12 | Sum $=70+21+69+44=204$ (1); Mean $=\frac{204}{4}=51$ (1). | 1+1 |
| 13 | Food $=\frac{35}{100}\times160=56$; Remaining share $=65\%$ (1); Everything else $=160-56=104$ (1). | 1+1 |
| 14 | Pulp $=\frac{24}{100}\times350=84$ mL (1); Remaining $=350-84=266$ mL (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Total of first group $=18\times64=1152$ (1); Total of second group $=12\times49=588$ (1); Combined average $=\frac{1740}{30}=58$ (1). | 1+1+1 |
| 16 | Sum of weekly averages $=135$ (1); Average $=\frac{135}{3}=45$ min per day (1); Every week has the same number of days, so each average has the same weight (1). | 1+1+1 |
| 17 | Weighted sum $=40\times2+85\times4=420$ (1); Sum of weights $=6$ (1); Weighted average $=70$ (1). | 1+1+1 |
| | **OR:** Total of first group $=25\times30=750$ (1); Total of second group $=20\times48=960$ (1); Combined average $=\frac{1710}{45}=38$ (1). | 1+1+1 |
| 18 | Sum of weekly averages $=132$ (1); Average $=\frac{132}{4}=33$ min per day (1); Every week has the same number of days, so each average has the same weight (1). | 1+1+1 |
| | **OR:** Weighted sum $=57\times4+91\times4=592$ (1); Sum of weights $=8$ (1); Weighted average $=74$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Sum of 5 observations $=5\times29=145$ (1); New sum $=145+59=204$ (2); New mean $=\frac{204}{6}=34$ (2). | 1+2+2 |
| | **OR:** Pure part in the first $=30\times\frac{22}{100}=6.6$ mL (1); Pure part in the second $=11.4$ mL (1); Total $=18$ mL in 90 mL (1); Concentration $=20\%$ (2). | 1+1+1+2 |
| 20 | Sum of 6 observations $=6\times47=282$ (1); New sum $=282+61=343$ (2); New mean $=\frac{343}{7}=49$ (2). | 1+2+2 |
| | **OR:** Pure part in the first $=20\times\frac{30}{100}=6$ mL (1); Pure part in the second $=9.6$ mL (1); Total $=15.6$ mL in 60 mL (1); Concentration $=26\%$ (2). | 1+1+1+2 |

**Mistakes to avoid:**

- Taking the combined average as $\frac{a_1+a_2}{2}$ when the groups have different sizes.

- Adding the percentages of two solutions to find the mixture, or averaging the two concentrations without using the volumes.
- Dividing the sum of weighted scores by the number of items instead of by the sum of the weights.
- Forgetting to change the volume when water is added: the pure amount stays the same but the total increases.
- Using a percentage as if it were an amount (e.g. $20\%$ of 5 L is 1 L, not 20 L).
- Reading a 100% stacked bar as if the bar lengths showed the actual totals; the actual totals are not shown.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND